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$$\|\vec{u} + \vec{v}\|^2 - \|\vec{u} - \vec{v}\|^2 = \langle \vec{u} + \vec{v}, \vec{u} + \vec{v} \rangle - \langle \vec{u} - \vec{v}, \vec{u} - \vec{v} \rangle =$$

$$= \langle \vec{u}, \vec{u} \rangle + \langle \vec{u}, \vec{v} \rangle + \langle \vec{v}, \vec{u} \rangle + \langle \vec{v}, \vec{v} \rangle - \langle \vec{u}, \vec{u} \rangle + \langle \vec{u}, -\vec{v} \rangle + \langle -\vec{v}, \vec{u} \rangle - \langle \vec{v}, \vec{v} \rangle =$$

$$= 2(\langle \vec{u}, \vec{v} \rangle + \langle \vec{v}, \vec{u} \rangle) = 2(\langle \vec{u}, \vec{v} \rangle + \overline{\langle \vec{u}, \vec{v} \rangle})$$

$$\Rightarrow \underline{a + \bar{a} = 2 \operatorname{Re}(a)}$$

$$\Rightarrow \|\vec{u} + \vec{v}\|^2 - \|\vec{u} - \vec{v}\|^2 = 4 \operatorname{Re}(\langle \vec{u}, \vec{v} \rangle)$$

$$\Rightarrow \operatorname{Re}(\langle \vec{u}, \vec{v} \rangle) = \frac{1}{4} (\|\vec{u} + \vec{v}\|^2 - \|\vec{u} - \vec{v}\|^2)$$

$$\Rightarrow \underline{\operatorname{Im}(a) = \operatorname{Re}(a \cdot (-i))}$$

$$\operatorname{Im}(\langle \vec{u}, \vec{v} \rangle) = \operatorname{Re}((-i)\langle \vec{u}, \vec{v} \rangle) = \operatorname{Re}(\langle \vec{u}, -i\vec{v} \rangle) =$$

$$= \frac{1}{4} (\|\vec{u} - i\vec{v}\|^2 - \|\vec{u} + i\vec{v}\|^2) \quad \blacksquare$$